

STA314, Lecture 6

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Alternative penalty methods: motivation

→ *coeff. in model selected by AIC*

An equivalent formulation of the **best subset selection** procedure with AIC is

$$\hat{b}^{AIC} = \operatorname{argmin}_{b \in \mathbb{R}^{p+1}} \left\{ \text{RSS}(b) + 2\hat{\sigma}^2 \#\{j = 2, \dots, p+1 : b_j \neq 0\} \right\}$$

Similarly, an equivalent formulation of the best subset selection with BIC is

$$\hat{b}^{BIC} = \operatorname{argmin}_{b \in \mathbb{R}^{p+1}} \left\{ \text{RSS}(b) + (\log n)\hat{\sigma}^2 \#\{j = 2, \dots, p+1 : b_j \neq 0\} \right\}$$

- ▶ The term $\#\{j = 2, \dots, p+1 : b_j \neq 0\}$ corresponds to the number of predictors used in the model. It can be viewed as measuring 'model complexity' (more predictors corresponds to more complex models).
- ▶ The two above minimization problems correspond to minimizing the sum of RSS (test error) and a penalty term that goes up with model complexity. This is called *regularization* or *penalized regression*.

There are other useful ways of measuring model complexity. The *lasso* and *ridge* penalties that follow are motivated by this idea.

The LASSO

Least Absolute Shrinkage and Selection Operator (Tibshirani 1996):

$$\hat{b}^L := \operatorname{argmin}_{b=(b_1, \dots, b_{p+1}) \in \mathbb{R}^{p+1}} \left\{ \text{RSS}(b) + \lambda \sum_{k=1}^p |b_{k+1}| \right\}.$$

Handwritten notes: Blue arrows point from the text "Least Absolute Shrinkage and Selection Operator" to the equation. A blue arrow points from the text "tuning parameter" to the λ in the equation. A red arrow points from the text "large" to the λ in the equation. A red arrow points from the text "usual linear model" to the $\lambda = 0$ in the equation.

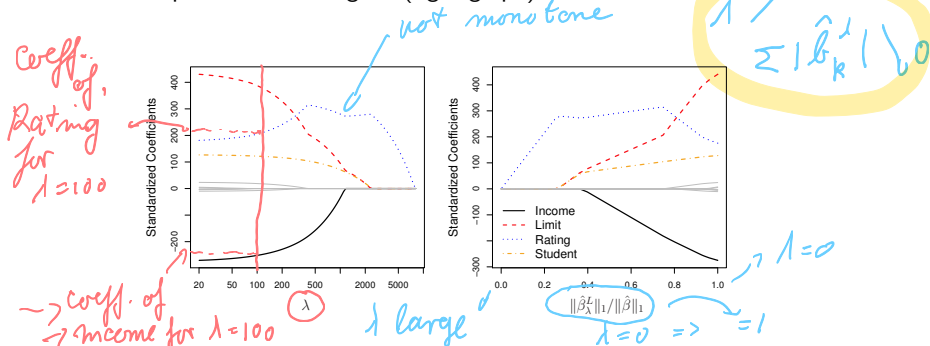
- ▶ The first part in above equation is the term we minimize when learning a linear model.
- ▶ λ is a tuning parameter which can be selected by cross-validation. In AIC this corresponded to $2\hat{\sigma}^2$, in BIC this corresponded to $(\log n)\hat{\sigma}^2$.
- ▶ The number of non-zero coefficients in AIC/BIC $\#\{k = 2, \dots, p+1 : b_j \neq 0\}$ is replaced by the sum of their absolute values $\sum_{k=1}^p |b_{k+1}|$.
- ▶ Penalty is minimized for $b_2 = \dots = b_{p+1} = 0$. $\lambda = 0$ is the usual linear model, "large" λ lead to all coefficients set to 0.
- ▶ Note: no penalty on the coefficient b_1 , usually we believe that there is a non-zero intercept.
- ▶ Not obvious, but one major advantage of this approach is computation. The problem is convex and can be solved with $p = 10000$ or even larger.

The lasso: example (Credit data set)

Least Absolute Shrinkage and Selection Operator (Tibshirani 1996):

$$\hat{b}^L := \operatorname{argmin}_{b=(b_1,\dots,b_{p+1}) \in \mathbb{R}^{p+1}} \left\{ RSS(b) + \lambda \sum_{k=1}^p |b_{k+1}| \right\}.$$

Example: plot of standardised regression coefficients against λ (left graph) and against relative norm of penalized and original (right graph).



Observation: for larger values of penalty, values of coefficients decrease. Closer look on next slide.

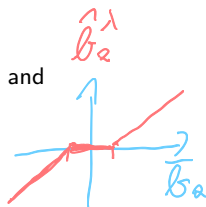
Why does the LASSO set coefficients to zero?

In general, there is no closed form solution for the LASSO optimization problem. To understand how it works, consider simple example (same as earlier in this lecture): one predictor with $\bar{x} = n^{-1} \sum_{i=1}^n x_i = 0$. Define

$$\overline{xy} := \frac{1}{n} \sum_{i=1}^n x_i y_i, \quad \overline{x^2} := \frac{1}{n} \sum_{i=1}^n x_i^2, \quad \bar{y} := \frac{1}{n} \sum_{i=1}^n y_i.$$

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linear regression

In that case, assuming $\overline{x^2} \neq 0$



and

$$\hat{\beta}_1^L = \bar{y}$$

$$\hat{\beta}_2 = \frac{\overline{xy}}{\overline{x^2}}$$

$$\hat{\beta}_2^L = \begin{cases} (\overline{xy} - .5\lambda/n)/\overline{x^2} & \overline{xy} > \lambda/2 \\ (\overline{xy} + .5\lambda/n)/\overline{x^2} & \overline{xy} < -\lambda/2 \\ 0 & n|\overline{xy}| \leq \lambda/2 \end{cases}$$

See detailed derivation in lectures.

$$\hat{\beta}_2 + \frac{1}{2n\overline{x^2}}$$

- ▶ solution of least squares without penalty in this case is $\hat{\beta}_2 = \overline{xy}/\overline{x^2}$.
- ▶ informal reason: "derivative" of absolute value function has jump at zero.

$$g(b_1, b_2) = \sum_{i=1}^n (y_i - b_1 - b_2 x_i)^2 + \lambda |b_2|$$



for fixed b_2 minimize over b_1 .

$$\frac{\partial}{\partial b_1} g(b_1, b_2) = - \underbrace{\sum_{i=1}^n 2(y_i - b_1 - b_2 x_i)}_{-2n(\bar{y} - b_1 - b_2 \bar{x})} + 0 \stackrel{!}{=} 0$$

$$\underbrace{-2n(\bar{y} - b_1 - b_2 \bar{x})}_{=0} = -2n(\bar{y} - b_1)$$

$$\Rightarrow b_1 = \bar{y} \text{ for any } b_2$$

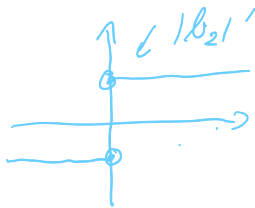
$$\Rightarrow \text{minimize } h(b_2) = \sum_{i=1}^n (y_i - \bar{y} - b_2 x_i)^2 + \lambda |b_2|$$

$$h'(b) = \begin{cases} \text{not defined} & , b_2 = 0 \\ -2 \sum_{i=1}^n x_i (y_i - \bar{y} - b_2 x_i) + \lambda & , b_2 > 0 \\ -\lambda & , b_2 < 0 \end{cases}$$

$$b_2 > 0 \Rightarrow |b_2| = b_2 \quad , \quad b_2 < 0 \Rightarrow |b_2| = -b_2$$

$$h'(b) = \begin{cases} \text{not defined} & , b_2 = 0 \\ -2 \sum_{i=1}^n x_i (y_i - \bar{y} - b_2 x_i) + 1 & , b_2 > 0 \\ -1 & , b_2 < 0 \end{cases}$$

$$= \begin{cases} \text{not def.} & , b_2 = 0 \\ -2n\bar{x}\bar{y} + 1 + 2n\bar{x}^2 b_2 & , b_2 > 0 \\ -2n\bar{x}\bar{y} - 1 + 2n\bar{x}^2 b_2 & , b_2 < 0 \end{cases}$$



Case 1 $|2n\bar{x}\bar{y}| < 1 \Rightarrow h$ minimized at $b_2 = 0$

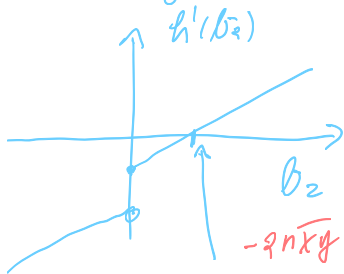


$$\underline{h'(b_2)} \begin{cases} \text{not def.} & , b_2 = 0 \\ > 0 & , b_2 > 0 \\ < 0 & , b_2 < 0 \end{cases} \Rightarrow$$

$h \searrow$ for $b_2 < 0$
 $h \nearrow$ for $b_2 > 0$

$$h'(b) = \begin{cases} \text{not def.} \\ -2n\overline{xy} + 1 + 2n\overline{x^2}b_2, & b_2 > 0 \leftarrow \\ -2n\overline{xy} - 1 + 2n\overline{x^2}b_2, & b_2 < 0 \end{cases}$$

Case 2: $2n\overline{xy} > 1$



$$-2n\overline{xy} + 1 + 2n\overline{x^2}b_2 = 0$$

$$h \text{ minimized at } b_2 = \frac{\overline{xy}}{\overline{x^2}} - \frac{1}{2n\overline{x^2}}$$

Comparison of LASSO, AIC, BIC in simple example

In setting of previous slide: $\hat{b}_1 = \hat{b}_1^L = \hat{b}_1^{AIC} = \hat{b}_1^{BIC} = \bar{y}$ and (derivation: Exercise)

$$\hat{b}_2^{AIC} = \begin{cases} \bar{xy}/\bar{x}^2 & , n(\bar{xy})^2/\bar{x}^2 > 2\hat{\sigma}^2 \\ 0 & , n(\bar{xy})^2/\bar{x}^2 \leq 2\hat{\sigma}^2 \end{cases} \quad \hat{b}_2^{BIC} = \begin{cases} \bar{xy}/\bar{x}^2 & , n(\bar{xy})^2/\bar{x}^2 > (\log n)\hat{\sigma}^2 \\ 0 & , n(\bar{xy})^2/\bar{x}^2 \leq (\log n)\hat{\sigma}^2 \end{cases}$$

For comparison: $\hat{b}_2 = \bar{xy}/\bar{x}^2$ and *lin. regression*

$$\hat{b}_2^L = \begin{cases} (\bar{xy} - .5\lambda/n)/\bar{x}^2 & n\bar{xy} > \lambda/2 \\ (\bar{xy} + .5\lambda/n)/\bar{x}^2 & n\bar{xy} < -\lambda/2 \\ 0 & n|\bar{xy}| \leq \lambda/2 \end{cases}$$

- ▶ If AIC or BIC selects predictor x into model, coefficient of that predictor is same as from full model
- ▶ Even if LASSO does not set coefficient of x to zero, the estimated coefficient $\hat{b}_2^L$ is smaller than in full model. The larger the penalty the smaller the estimated coefficient.

Comments on LASSO

- ▶ In general, there is no simple closed form solution for LASSO.
- ▶ Under mild assumptions on the vectors X_i , the lasso problem is *strictly convex*. This implies the existence of a unique minimizer which is easy to compute using modern optimization procedures.
- ▶ Even with the R implementation, LASSO solutions can be computed with several *thousands* of predictors fairly efficiently.

LASSO vs best subset selection methods with AIC, BIC and cross-validation

- + LASSO has substantial advantages in terms of computational complexity. Can be computed even for very large numbers of predictors.
- = Similarly to BIC, LASSO can perform consistent model selection (under certain technical assumptions).
- + Because of penalization, LASSO usually introduces bias. However, penalty can also help to reduce variance (bias-variance tradeoff).
- ▶ In practice, after running a LASSO to select variables, it is common to **refit model using only the selected variables**.
- ▶ To make sure that all predictors get same treatment, it is advisable to standardise all predictors to sample mean zero and sample standard deviation 1.

Some useful functions in R

Library `glmnet` contains functions for LASSO and ridge regression.

Library `plotmo` contains useful plotting functions for output of lasso and ridge.

`lasso.mod = glmnet(x,y,alpha = 1, lambda = grid)` runs lasso regression.

- ▶ Inputs: `x` output of function `model.matrix`, `y` response values, `alpha = 1` means do lasso, `lambda = grid` means use this grid of `lambda` values to compute lasso estimators.
- ▶ Output: object, can be used for plotting or extracting coefficients.
- ▶ `coef(lasso.mod)` provides array with estimated coefficients (each row corresponds to one `lambda` value from `grid`.)
- ▶ Plot result using function

```
plot_glmnet(lasso.mod)
```

`cv.la = cv.glmnet(x,y,alpha = 1, lambda = grid)` can be used for cross-validation. Access to `lambda` selected by cross-validation and by one standard error rule through

```
cv.la$lambda.min, cv.la$lambda.1se
```

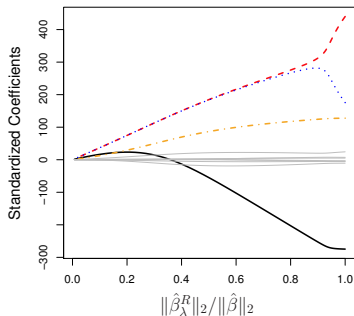
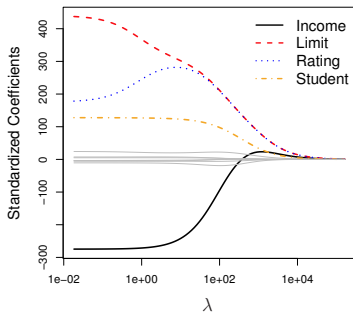
See R code example: [STA314-Lecture06-HittersDataAnalysis-LASSO](#)

Ridge regression

$$\hat{b}^R := \operatorname{argmin}_{b=(b_1, \dots, b_{p+1}) \in \mathbb{R}^{p+1}} \sum_{i=1}^n (Y_i - X_i^\top b)^2 + \lambda \sum_{k=1}^p b_{k+1}^2.$$

for LASSO was $\sum |b_{k+1}|$

- ▶ Instead of looking at the sum of absolute values of b_{k+1} we look at the sum of their squares.
- ▶ λ is a tuning parameter that can be selected by cross-validation.
- ▶ Ridge regression does not set coefficients to zero!
- ▶ Motivation for ridge regression: introduce some bias hoping to reduce variance.



Ridge regression does not set coefficients to zero

Same setting as before for LASSO.

for
Ridge

```
> coef(ridge.mod)[,100] # lambda = 0.01
(Intercept)      Age      Education      Limit
-198.8070412    -2.2934420     1.8760364     0.1734225

> coef(ridge.mod)[,80] # lambda = 10
(Intercept)      Age      Education      Limit
-195.1052213    -2.2666645     1.8472106     0.1724079

> coef(ridge.mod)[,70] # lambda = 43
(Intercept)      Age      Education      Limit
-142.6960719    -1.9079664     1.4656002     0.1582081

> coef(ridge.mod)[,1] # lambda = 10^10
(Intercept)      Age      Education      Limit
5.200150e+02    2.245932e-09  -5.445753e-08  7.881306e-09
```

How does ridge regression work? A special case.

$\lambda \geq 0$

Simple example: one predictor with $\bar{x} = n^{-1} \sum_{i=1}^n x_i = 0$. Define

$$\overline{xy} := \frac{1}{n} \sum_{i=1}^n x_i y_i, \quad \overline{x^2} := \frac{1}{n} \sum_{i=1}^n x_i^2, \quad \bar{y} := \frac{1}{n} \sum_{i=1}^n y_i.$$

Then $\hat{b}_1^R = \bar{y}$ and (Exercise)

if $\overline{x^2} \neq 0$

$$\hat{b}_2^R = \frac{n\overline{xy}}{n\overline{x^2} + \lambda} = \hat{b}_2 \frac{n\overline{x^2}}{n\overline{x^2} + \lambda}$$

from linear regression
 < 1 if $\lambda > 0$

- ▶ Ridge does not give values of exactly zero, so does not perform model selection.
- ▶ Larger values are pushed to zero. The larger the value the stronger the effect.

Recap: eigenvalues and eigenvectors.

Throughout this slide let M denote a symmetric $k \times k$ matrix with real entries.

A vector $v \in \mathbb{R}^k$ is *eigenvector* of M with *eigenvalue* α iff $Mv = \alpha v$.

$v \neq 0$

Theorem For every symmetric matrix M we can find an orthonormal basis of $\mathbb{R}^k$ which is spanned by eigenvectors of M and all eigenvalues are real.

Theorem M is invertible if and only if all eigenvalues of M are non-zero.

Exercise: if M invertible and v eigenvector of M with eigenvalue α then v is eigenvector of M^{-1} with eigenvalue $1/\alpha$.

Exercise: if $M = A^T A$ for a matrix A the all eigenvalues of M are non-negative.

Ridge regression: general case.

The general case: assume predictors are d -dimensional and for $j = 1, \dots, d$ we have $\sum_{i=1}^n x_{i,j} = 0$. Let $y_i = b_1 + \tilde{b}^\top \tilde{X}_i + \varepsilon_i$ and define for $\tilde{X}_i = (x_{i,1}, \dots, x_{i,d})^\top$

$$\tilde{X} = \begin{pmatrix} x_{1,1}, \dots, x_{1,d} \\ \vdots \\ x_{n,1}, \dots, x_{n,d} \end{pmatrix}, \quad Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

recall:
lin
regression

One can prove (see notes for detailed derivation): $\hat{b}_1^R = \bar{y}$ and

$$\check{b}^R := (\hat{b}_2^R, \dots, \hat{b}_{d+1}^R)^\top = (\tilde{X}^\top \tilde{X} + \lambda I_d)^{-1} \tilde{X}^\top Y$$

$(\tilde{X}^\top \tilde{X})^{-1} \tilde{X}^\top Y$
assume exists

where I_d is $d \times d$ identity matrix.

- ▶ Even if the matrix $\tilde{X}^\top \tilde{X}$ is not invertible, the matrix $\tilde{X}^\top \tilde{X} + \lambda I_d$ is invertible if $\lambda > 0$.
- ▶ One can prove: if $\tilde{X}^\top \tilde{X}$ has full rank then $\text{Var}(w^\top \check{b}) \leq \text{Var}(w^\top \check{b}^R)$ for any $w \neq 0$ if $\lambda > 0$. So ridge regression reduces variance of predictions.
- ▶ For $\lambda \neq 0, \check{b} \neq 0$ ridge introduces bias

$$\mathbb{E}[\check{b}^R] - \tilde{b} = [(\tilde{X}^\top \tilde{X} + \lambda I_d)^{-1} \tilde{X}^\top \tilde{X} - I_d] \tilde{b} \neq 0.$$

$$h(\beta_1, \beta) = \sum_{i=1}^n (y_i - \beta_1 - \beta^T \tilde{x}_i)^2 + \lambda \sum_{k=2}^{d+1} \beta_k^2$$

Exercise: minimizing over β_1 gives

$$\hat{\beta}_1 = \bar{y} \text{ for any } \beta, \text{ with: } \sum_{i=1}^n x_{i,j} = 0$$

remains to minimize

$$g(\beta_2, \dots, \beta_{d+1}) = \sum_{i=1}^n (y_i - \bar{y} - \beta^T \tilde{x}_i)^2 + \lambda \sum_{k=2}^{d+1} \beta_k^2$$

$$\frac{\partial g}{\partial \beta_2} = -2 \sum_{i=1}^n x_{i,2} (y_i - \bar{y} - \beta^T \tilde{x}_i) + 2\lambda \beta_2 \stackrel{!}{=} 0$$

$$\vdots$$

$$\frac{\partial g}{\partial \beta_{d+1}} = -2 \sum_{i=1}^n x_{i,d+1} (y_i - \bar{y} - \beta^T \tilde{x}_i) + 2\lambda \beta_{d+1} \stackrel{!}{=} 0$$

matrix form:

$$\begin{aligned} & \text{(Exercise)} \quad -2 \tilde{X}^T Y - 2 \tilde{X}^T \tilde{X} \beta + 2\lambda I_{d \times d} \beta \stackrel{!}{=} 0 \\ & \Leftrightarrow (\tilde{X}^T \tilde{X} + \lambda I_{d \times d}) \beta = \tilde{X}^T Y \end{aligned}$$

$$\Leftrightarrow \underbrace{(\tilde{X}^T \tilde{X} + I_{d \times d})}_{\text{invertible}} \tilde{G} = \tilde{X}^T Y$$

invertible:

$d_1, \dots, d_d$ eigenvalues of $\tilde{X}^T \tilde{X}$ eigenvectors: $V_1, \dots, V_d$

$$\Rightarrow d_1 \geq 0, \dots, d_d \geq 0$$

$$(\tilde{X}^T \tilde{X} + I_{d \times d}) V_k = \tilde{X}^T \tilde{X} V_k + I_{d \times d} V_k$$

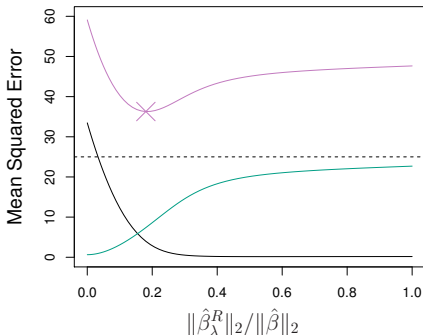
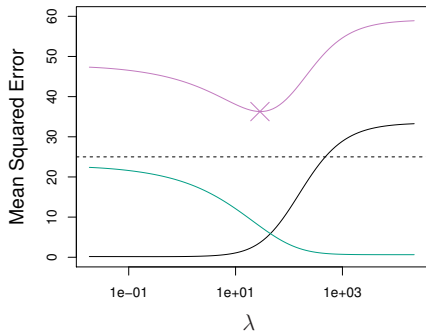
$$= d_k V_k + 1 V_k = (d_k + 1) V_k$$

$$\Rightarrow \tilde{X}^T \tilde{X} + I_{d \times d} \text{ has eigenvalues } \underbrace{d_k}_{\geq 0} + \underbrace{1}_{> 0} > 0$$

$\hookrightarrow$ invertible since no zero eigenvalues

Ridge regression: why is it helpful?

Example of squared bias (black), variance (green) and test MSE (purple) of ridge regression in one simulated example. Dashed horizontal line is irreducible part of error.



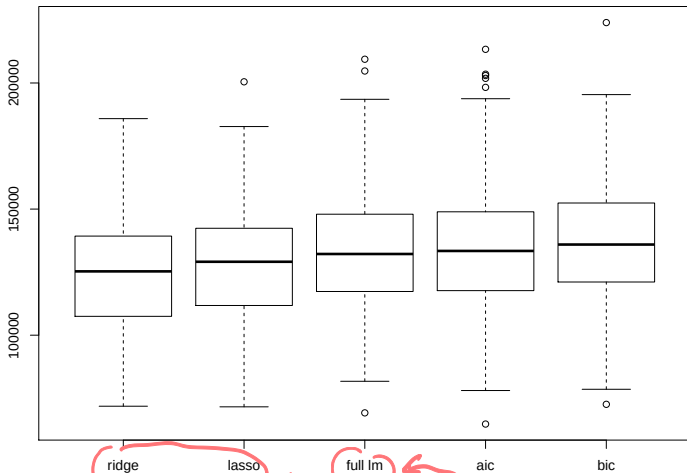
- ▶ $\lambda = 0$ or equivalently $\|\hat{\beta}_\lambda^R\|_2 / \|\hat{\beta}\|_2 = 1$ corresponds to least squares solution.
- ▶ Best MSE for ridge is better compared to MSE of least squares, this corresponds to optimal bias-variance trade-off.

Ridge, LASSO, best subset selection: comments on comparison

- ▶ Ridge regression is helpful for prediction if there are many predictors that are useful but 'weak', and if there is high correlation between some of the predictors.
- ▶ Ridge tends to give predictors which are highly correlated similar regression coefficients (example: `Limit` and `Rating` in `Credit` data).
- ▶ LASSO works best if only few predictors are important and many predictors have no influence on outcome ('sparse' case).
- ▶ LASSO, AIC, BIC can be used to perform model selection (recall: AIC does not perform consistent model selection while BIC and LASSO can). Ridge regression does not perform model selection.
- ▶ If there are many predictors, LASSO and Ridge are computationally less expensive compared to best subset selection with AIC, BIC or cross-validation.

Ridge, LASSO, best subset selection: comparison for Hitters data

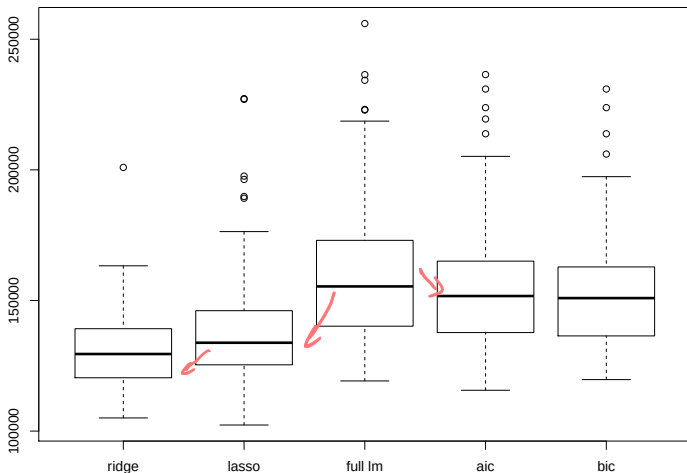
Test error for various methods based on 50-50 split of data set, corresponds to 131 training data. Ridge a bit better compared to lasso, lasso a bit better compared to full model. BIC a bit worse compared to full model.



a bit better than

Ridge, LASSO, best subset selection: comparison for Hitters data

Test error for various methods with roughly 70 training data. Ridge and lasso show more substantial advantage over linear model, AIC and BIC a bit better compared to linear model. Lesson: results depend on sample size. In smaller samples regularization is more helpful.



Some R examples.

Ridge and other examples: `STA314-Lecture06-RExamples2`

Simulation-based comparisons: `STA314-Lecture-06-Simulations`